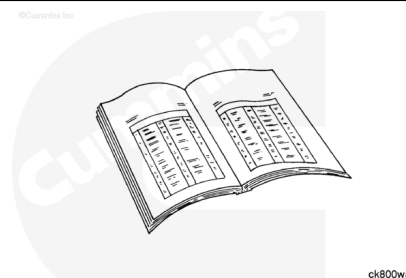


Trainings ▾

General Information

This procedure is for front gear train engines **only** in which the camshaft has already been removed. Refer to Procedure 001-008 (/qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html).

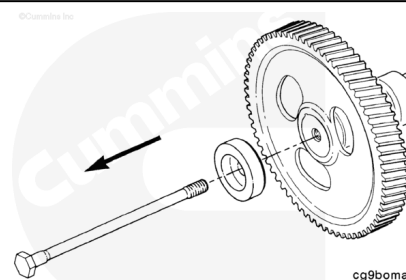
Note : For rear gear train engines, the camshaft gear **must** be removed to remove the camshaft. Refer to Procedure 001-012 (/qs3/pubsys2/xml/en/procedures/40/40-001-012-tr.html).



Remove

On bolted camshafts, remove the camshaft bolt and washer.

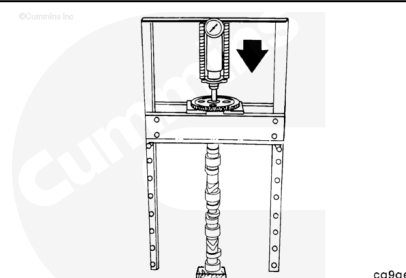
Note : Bolted camshafts are **only** used on 1991 engines with an inline injection pump.



The camshaft gear can be removed from the camshaft by the following methods:

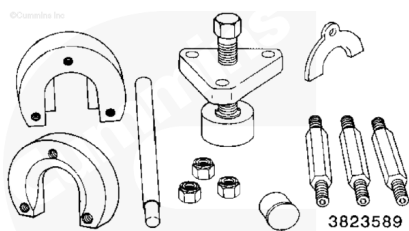
⚠ CAUTION ⚠

The camshaft gear must be fully supported when using a hydraulic press. Failure to properly support the camshaft gear will result in damage to the camshaft gear.

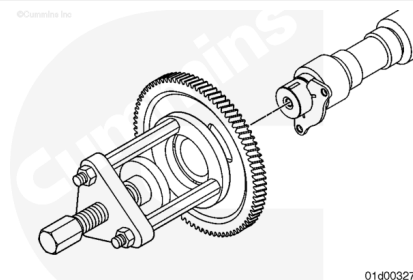


Use a hydraulic press

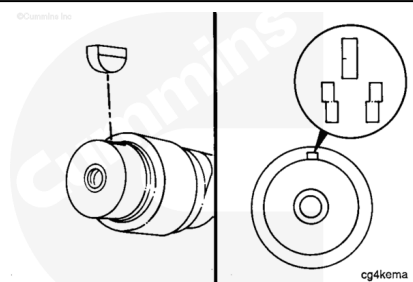
Use the camshaft gear removal and installation tool, Part Number 3823589.



Remove the camshaft gear from the nose of the camshaft.



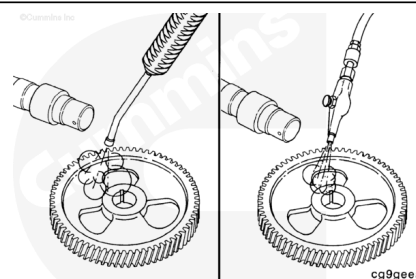
Remove the camshaft gear locating key or dowel pin.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the camshaft gear.

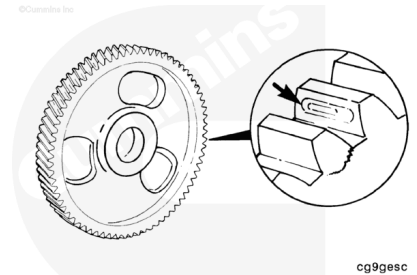
Dry with compressed air.

Inspect the camshaft gear teeth for pitting. Look for cracks at the root of the teeth.

Inspect the camshaft gear teeth for discoloration or signs of excessive heat (bluing).

If any damage is present, replace the camshaft gear.

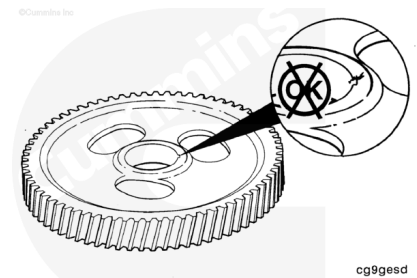
Note : If there is damage to the camshaft gear teeth or there are signs of excessive heat, make sure to inspect the associated crankshaft and fuel pump gears for damage. Measure camshaft gear backlash upon installation.



Note : If the fretting, burrs, or raised material burrs can not be removed, with abrasive pad, Part Number 3823258, replace camshaft gear.

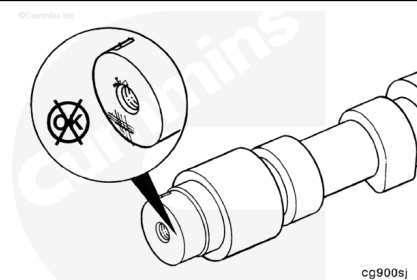
Inspect the camshaft bore for fretting or burrs.

Inspect the camshaft gear for burrs.



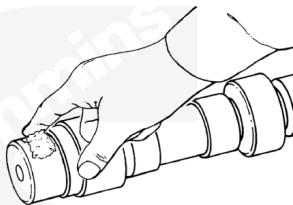
Inspect the camshaft nose for frets or burrs.

Note : If frets or burrs can not be removed with an abrasive pad, Part Number 3823258, or equivalent, replace the camshaft.



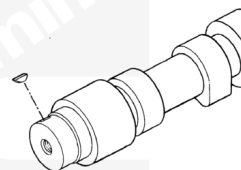
Install

Lubricate the camshaft nose with Lubriplate™ 105, or equivalent.



cg900wc

Install the camshaft gear locating key or dowel pin with a plastic mallet.

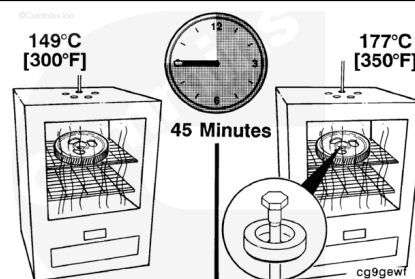


cg9kema

The camshaft gear can be installed by using the following methods:

⚠ CAUTION ⚠

The camshaft gear will be permanently distorted if overheated. The oven temperature must never exceed, for the given camshaft gear type, the temperature specification given below.



Non-Bolted Camshaft Gears (1991 Engines only)

Heat the camshaft gear to 149°C [300°F] for 45 minutes.

All Other Camshaft Gears

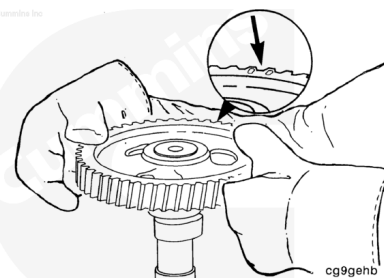
Heat the camshaft gear to 177°C [350°F] for 45 minutes.

⚠ WARNING ⚠

To prevent personal injury, wear protective gloves to handle the hot gear.

Install the camshaft gear with the timing marks away from the camshaft.

Install the camshaft gear onto the nose of the camshaft. Align the camshaft gear keyway with the camshaft locating key/dowel pin.

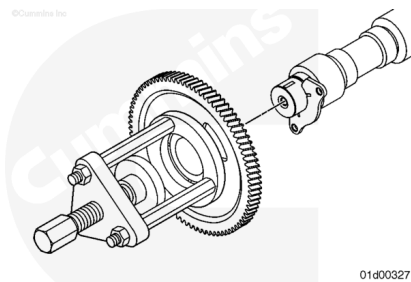


cg9gehb

Using the camshaft gear removal and installation tool, Part Number 3823589.

Note : Install the camshaft gear with the timing marks away from the camshaft.

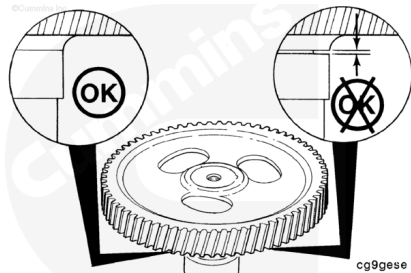
Install the camshaft gear onto the nose of the camshaft. Align the camshaft gear keyway with the camshaft locating key/dowel pin.



⚠ CAUTION ⚠

Make sure the gear is seated against camshaft shoulder.

Use a 0.025 mm [0.001 inch] feeler gauge to see if the feeler gauge can be inserted between the camshaft gear and the shoulder on the camshaft. If the feeler gauge can be inserted, the camshaft gear is **not** properly seated.



For bolted camshaft gears, install the camshaft bolt and washer.

Tighten the bolt.

Torque Value:	Step 1	27 n.m [20 ft-lb]
	Step 2	Rotate capscrew 180 degrees.

